```
import tensorflow as tf
import numpy as np
import matplotlib.pyplot as plt
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Flatten, Dense

(x_train, y_train), (x_test, y_test) = tf.keras.datasets.mnist.load_data()
```

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz
11490434/11490434 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
```

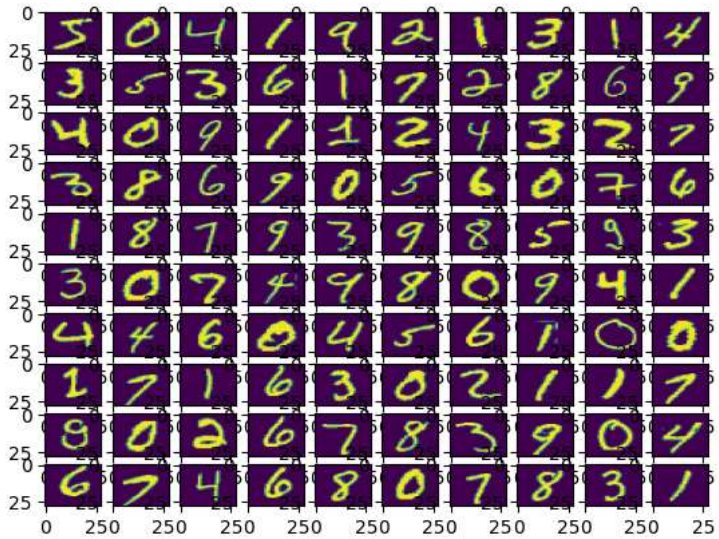
```
gray_scale = 255

x_train = x_train.astype('float32') / gray_scale
x_test = x_test.astype('float32') / gray_scale

print("Feature matrix (x_train):", x_train.shape)
print("Target matrix (y_train):", y_train.shape)
print("Feature matrix (x_test):", x_test.shape)
print("Target matrix (y_test):", y_test.shape)
```

```
Feature matrix (x_train): (60000, 28, 28)
Target matrix (y_train): (60000,)
Feature matrix (x_test): (10000, 28, 28)
Target matrix (y_test): (10000,)
```

```
fig, ax = plt.subplots(10, 10)
k = 0
for i in range(10):
    for j in range(10):
        ax[i][j].imshow(x_train[k].reshape(28, 28), aspect='auto')
        k += 1
plt.show()
```



```
model = Sequential([
    Flatten(input_shape=(28, 28)),
    Dense(256, activation='sigmoid'),
    Dense(128, activation='sigmoid'),
    Dense(10, activation='softmax'),
])
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/reshaping/flatten.py:37: UserWarning: Do not pass an `input_shape`
  super().__init__(**kwargs)
```

```
model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])
```

```
mod = model.fit(x_train, y_train, epochs=10,
          batch_size=2000,
          validation_split=0.2)

print(mod)
```

```
Epoch 1/10
24/24 ━━━━━━━━━━━━━━━━━━━━ 3s 81ms/step - accuracy: 0.2217 - loss: 2.2858 - val_accuracy: 0.5845 - val_loss: 1.8044
Epoch 2/10
24/24 ━━━━━━━━━━━━━━━━━━━━ 2s 67ms/step - accuracy: 0.6697 - loss: 1.6411 - val_accuracy: 0.8142 - val_loss: 1.1259
Epoch 3/10
24/24 ━━━━━━━━━━━━━━━━━━━━ 2s 50ms/step - accuracy: 0.8131 - loss: 1.0285 - val_accuracy: 0.8618 - val_loss: 0.7281
Epoch 4/10
24/24 ━━━━━━━━━━━━━━━━━━━━ 1s 49ms/step - accuracy: 0.8534 - loss: 0.6991 - val_accuracy: 0.8857 - val_loss: 0.5333
Epoch 5/10
24/24 ━━━━━━━━━━━━━━━━━━━━ 2s 74ms/step - accuracy: 0.8802 - loss: 0.5302 - val_accuracy: 0.8986 - val_loss: 0.4322
Epoch 6/10
24/24 ━━━━━━━━━━━━━━━━━━━━ 2s 75ms/step - accuracy: 0.8917 - loss: 0.4390 - val_accuracy: 0.9062 - val_loss: 0.3739
Epoch 7/10
24/24 ━━━━━━━━━━━━━━━━━━━━ 1s 50ms/step - accuracy: 0.9027 - loss: 0.3799 - val_accuracy: 0.9102 - val_loss: 0.3360
Epoch 8/10
24/24 ━━━━━━━━━━━━━━━━━━━━ 1s 48ms/step - accuracy: 0.9090 - loss: 0.3435 - val_accuracy: 0.9164 - val_loss: 0.3092
Epoch 9/10
24/24 ━━━━━━━━━━━━━━━━━━━━ 1s 47ms/step - accuracy: 0.9157 - loss: 0.3122 - val_accuracy: 0.9213 - val_loss: 0.2888
Epoch 10/10
24/24 ━━━━━━━━━━━━━━━━━━━━ 1s 47ms/step - accuracy: 0.9184 - loss: 0.2977 - val_accuracy: 0.9248 - val_loss: 0.2729
<keras.src.callbacks.history.History object at 0x7eb2e56db2c0>
```

```
results = model.evaluate(x_test, y_test, verbose=0)
print('Test loss, Test accuracy:', results)
```

```
Test loss, Test accuracy: [0.2781641483306885, 0.921999990940094]
```

```
plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.plot(mod.history['accuracy'], label='Training Accuracy', color='blue')
plt.plot(mod.history['val_accuracy'],
         label='Validation Accuracy', color='orange')
plt.title('Training and Validation Accuracy', fontsize=14)
plt.xlabel('Epochs', fontsize=12)
plt.ylabel('Accuracy', fontsize=12)
plt.legend()
plt.grid(True)

plt.subplot(1, 2, 2)
plt.plot(mod.history['loss'], label='Training Loss', color='blue')
plt.plot(mod.history['val_loss'], label='Validation Loss', color='orange')
plt.title('Training and Validation Loss', fontsize=14)
plt.xlabel('Epochs', fontsize=12)
plt.ylabel('Loss', fontsize=12)
plt.legend()
plt.grid(True)

plt.suptitle("Model Training Performance", fontsize=16)
plt.tight_layout()
plt.show()
```

Model Training Performance
Training and Validation Accuracy
Accuracy
Epochs
Training Accuracy
Validation Accuracy
Training and Validati
Loss
Epochs